



one signifies the laser spot size at target that is limited only by the laws of diffraction. For a real laser beam, beam quality is greater than one and hence the focused spot sizes are larger than the diffraction limited values.

Maintaining good laser beam quality as the power is scaled up is one of the most complex and challenging tasks in the laser system design. Large telescope aperture size is preferred as it produces tighter focusing and, hence, increased laser power densities at the target.

Low atmospheric transmission losses are important as the laser beam has to propagate over

laser, free-electron lasers, solid-state lasers and fibre lasers.

Gas dynamic lasers

Gas dynamic lasers usually use a combustion chamber, supersonic expansion nozzle and CO_2 , in a mixture with nitrogen or helium, as the laser medium. Different constituents of a combustion-driven gas dynamic laser are shown in Fig. 7. The gas dynamic laser derives its energy from combustion of a suitable fuel-oxidiser mixture, which means that it does not require any electrical energy for its operation.

A gas dynamic laser achieves population inversion by rapid

(30 torr) as compared to that of other similar HPL sources, well-established technology for direct atmospheric exhaust of lasing gases, well-suited for platform mounting. Airborne Laser Lab program of the USA launched in 1976 used gas dynamic CO_2 laser as the HPL source.

Chemical lasers

A chemical laser derives the energy required to produce population inversion and consequent laser emission from a chemical reaction. Importance of chemical lasers lies in their ability to generate continuous wave output power reaching several megawatts. Besides, the range of wavelengths generated by these lasers is well absorbed by metals, which makes these suitable for cutting and drilling operations in the industry, and more so as HPL sources for DEW applications.

There are two categories of chemical lasers. One category has HF and DF, and the other has the two iodine lasers including chemical oxy-iodine laser and all gas-phase iodine laser (AGIL). Both HF/DF and chemical oxy-iodine laser systems have demonstrated their capability to be operated at megawatt-class continuous-wave output power. Lasing action has been demonstrated in AGIL and efforts are on to scale the output to higher power levels.

Lasing wavelength of 3.8 microns in the case of DF laser has good atmospheric transmission as compared to that of HF laser emitting at 2.7 microns wavelength. HF laser is, however, better suited to Space based applications. Both have high specific energy and relatively smaller size as compared to gas dynamic lasers. DF laser of MIRACL system has demonstrated power level of 2.2 megawatts. However, HF/DF lasers produce highly toxic and explosive gases during operation and also require complex logistics.

Chemical oxy-iodine laser is a low-temperature, low-pressure device

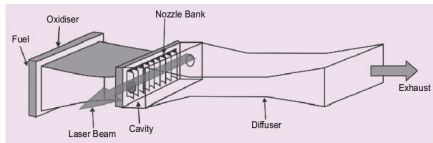


Fig. 7: Functional schematic of a combustion-driven gas dynamic laser

long atmospheric paths. For this to happen, one of the most important requirements is that the laser wavelength falls within the available transmission windows of the atmosphere in 0.4 micron to 1.7 microns (visible - NIR), three microns to five microns (MWIR) and 8 microns to 14 microns (FWIR) bands.

In addition to these requirements, another important criterion for HPL is the need to adopt power-scalable laser design architecture. The basic physics and technology of the laser system design should allow power scaling to megawatt power level needed for long-range high-energy laser weapons.

Not all laser sources, in general, meet the above requirements. Elite categories of laser sources for DEW applications are gas dynamic CO_2 lasers, chemical lasers including HF/DF laser and chemical oxy-iodine

expansion of high-temperature high-pressure laser gas mixture produced during combustion to a near vacuum in an adiabatic process through an integrated supersonic nozzle bank. This type of laser, which is also capable of producing hundreds of kilowatts of CW power at 10.6 microns initially assumed importance as a potential HPL weapon, but it seems to have lost out in the race in favour of HF/DF and chemical oxy-iodine laser systems due to heavy absorption of 10.6 microns wavelength by water vapour in the atmosphere.

Salient features of a gas dynamic laser include long-lasing wavelength of 10.6 microns, larger beam divergence, larger telescope aperture size requirements, proven technology and non-toxic, power scale-up proven up to megawatt power level, relatively high cavity pressure